

GGL90 Validation Report

MITgcm vs Python Port Comparison (single-step replay)

Generated: 2026-09-27 05:56:38

MITgcm Reference:

File: mitgcm_ggl90_outputs_isomip_12.nc
UUID: c716d756-1f0d-4ab3-a7b5-d36c2ffa10e9

Python Port Output:

File: python_ggl90_outputs_isomip_12.nc
UUID: c716d756-1f0d-4ab3-a7b5-d36c2ffa10e9

Provenance: ☐ UUIDs match

Timesteps: 12
Grid: 50 × 100 × 30

Fields compared: visc_az, diff_kz, mixing_length, tke_after
(GGL90 single-step replay: Python fed MITgcm's own captured
tke_before(t); only the one-step update is compared, isolating
agreement on the physics formula from trajectory drift.)

GGL90 Field Statistics

Wet-cell comparison (NaN-excluded below-seafloor/above-ice-shelf cells)

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visc_az  (Vertical eddy viscosity (K_m), m^2/s)
          Status:  EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 0.0000e+00  Max |diff|: 2.3898e-10
Median rel:   0.0000e+00  P95 rel:   0.0000e+00  Max rel:   2.1295e-07
          Fraction >1% rel error: 0.0000%
Points compared: 1437204 (excluded 362796 NaN cells)

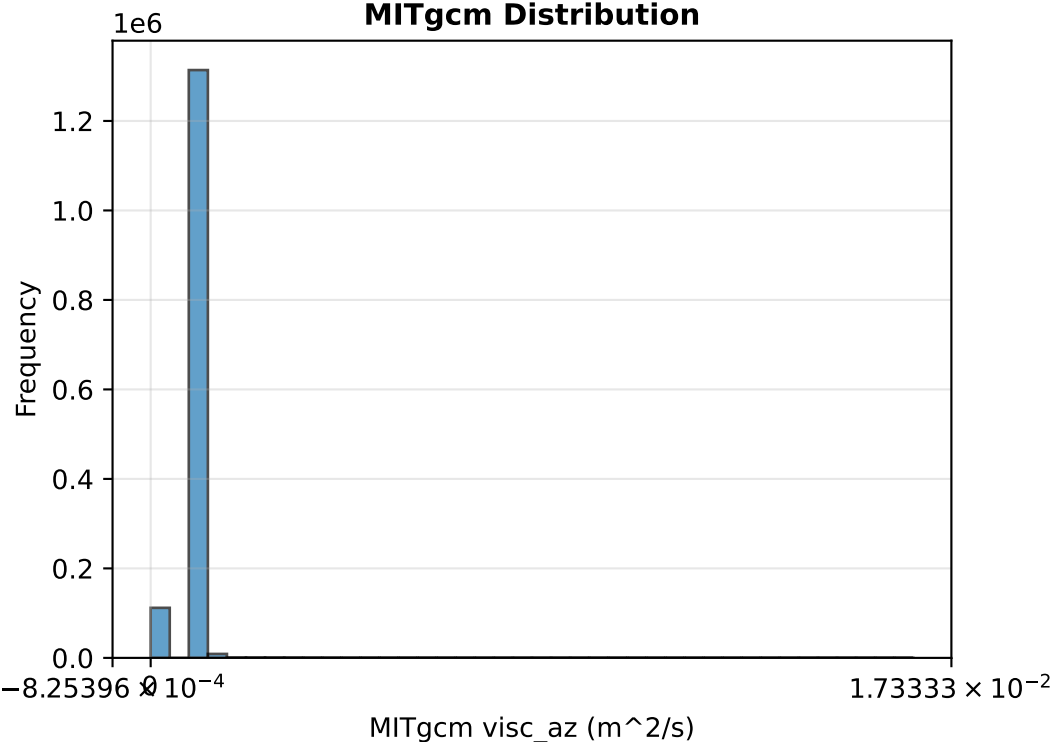
diff_kz  (Vertical eddy diffusivity (K_h), m^2/s)
          Status:  EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 0.0000e+00  Max |diff|: 2.9049e-03
Median rel:   0.0000e+00  P95 rel:   0.0000e+00  Max rel:   3.0000e+00
          Fraction >1% rel error: 0.0770%
Points compared: 1437204 (excluded 362796 NaN cells)

mixing_length (GGL90 mixing length (L), m)
          Status:  EXCELLENT
Median |diff|: 4.9873e-18  P95 |diff|: 2.1316e-14  Max |diff|: 4.2817e-03
Median rel:   3.5691e-15  P95 rel:   6.4581e-12  Max rel:   3.2098e-04
          Fraction >1% rel error: 0.0000%
Points compared: 1437204 (excluded 362796 NaN cells)

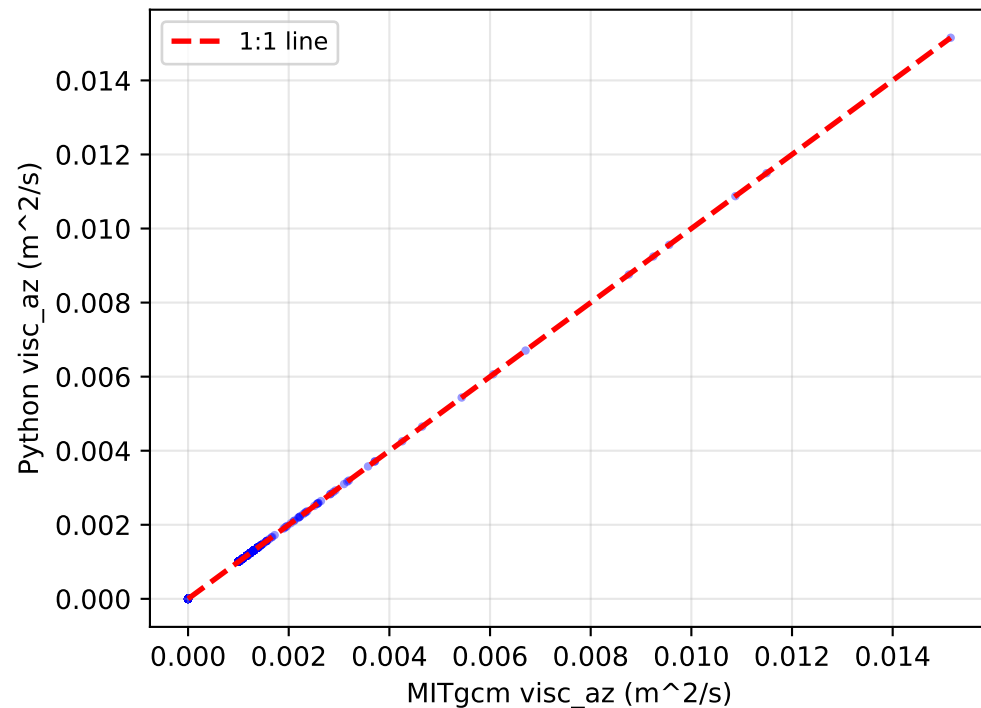
tke_after (TKE at t+1 (single-step replay), m^2/s^2)
          Status:  EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 6.3151e-17  Max |diff|: 9.0811e-06
Median rel:   0.0000e+00  P95 rel:   1.3664e-06  Max rel:   2.3269e+00
          Fraction >1% rel error: 1.2478%
Points compared: 1437204 (excluded 362796 NaN cells)
```

visc_az (Vertical eddy viscosity (K_m)) -- MITgcm vs Python

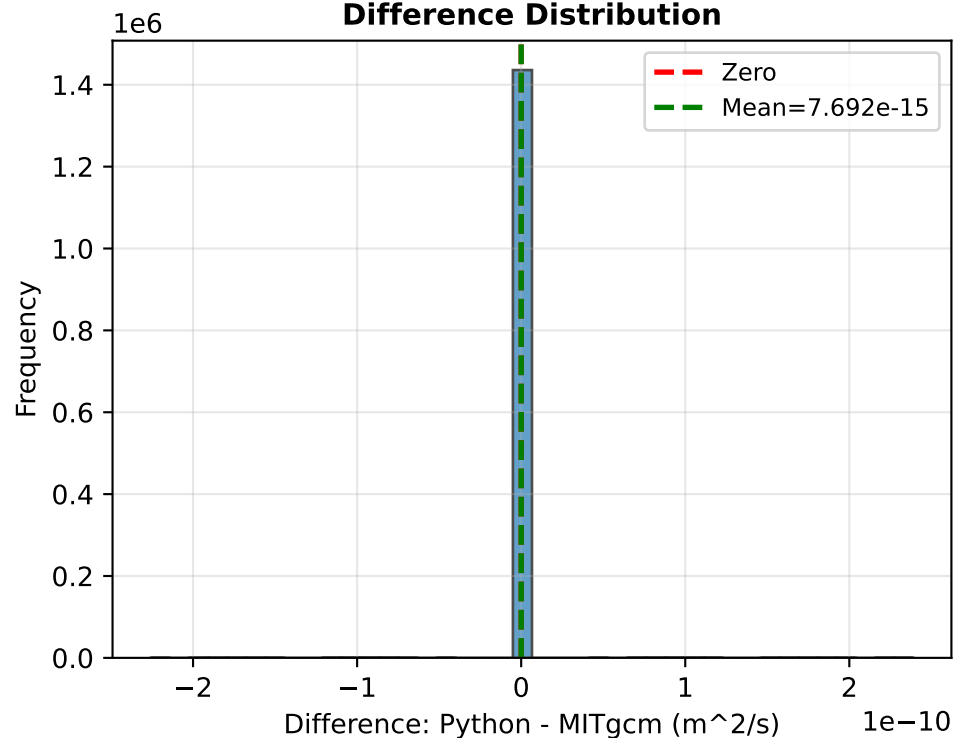
MITgcm Distribution



Scatter Plot (20000 of 1437204 pts)



Difference Distribution



visc_az Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 0.0000e+00
Max |diff|: 2.3898e-10

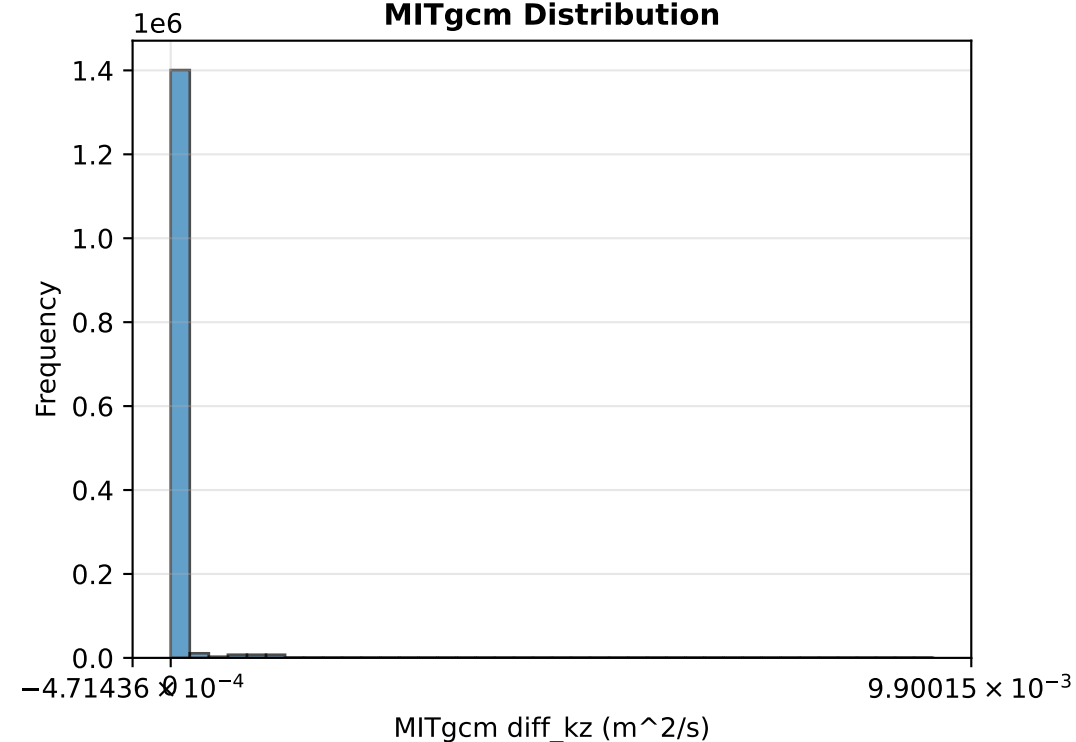
Median rel: 0.0000e+00
P95 rel: 0.0000e+00
Max rel: 2.1295e-07

Fraction >1% rel: 0.0000%
Points compared: 1437204
NaN-excluded: 362796

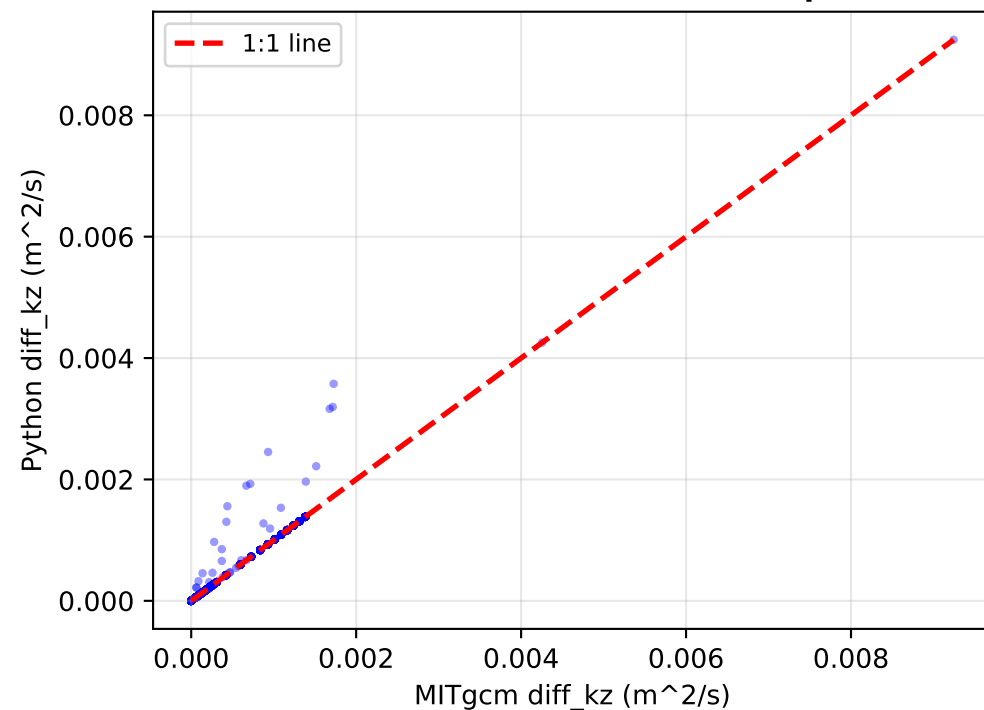
Assessment:
EXCELLENT

diff_kz (Vertical eddy diffusivity (K_h)) -- MITgcm vs Python

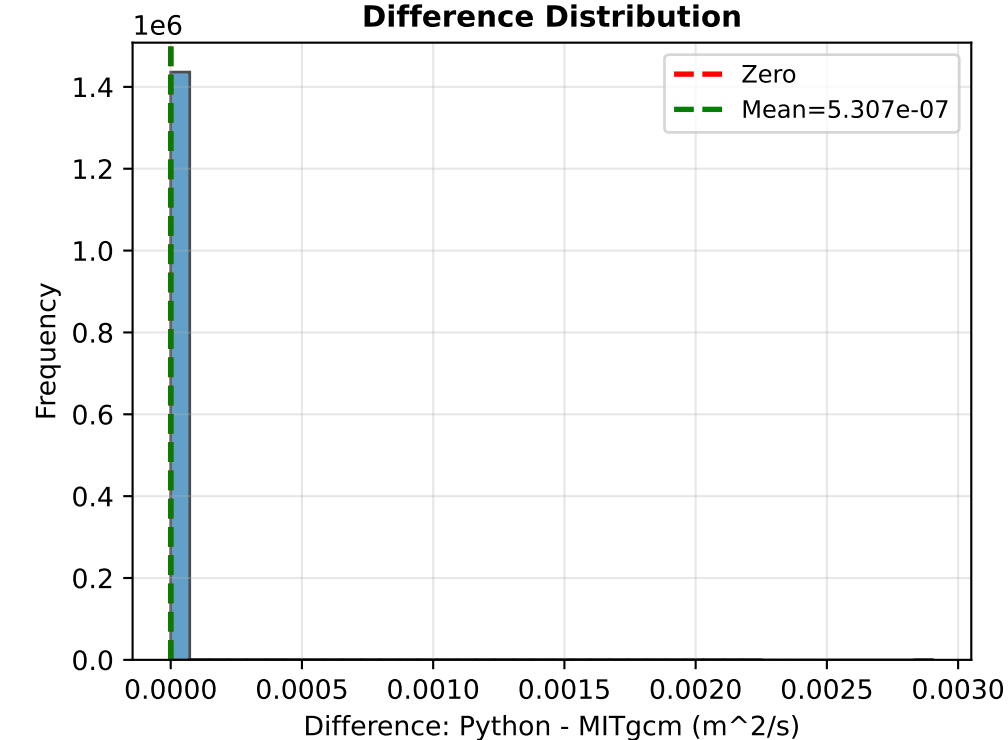
MITgcm Distribution



Scatter Plot (20000 of 1437204 pts)



Difference Distribution



diff_kz Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 0.0000e+00
Max |diff|: 2.9049e-03

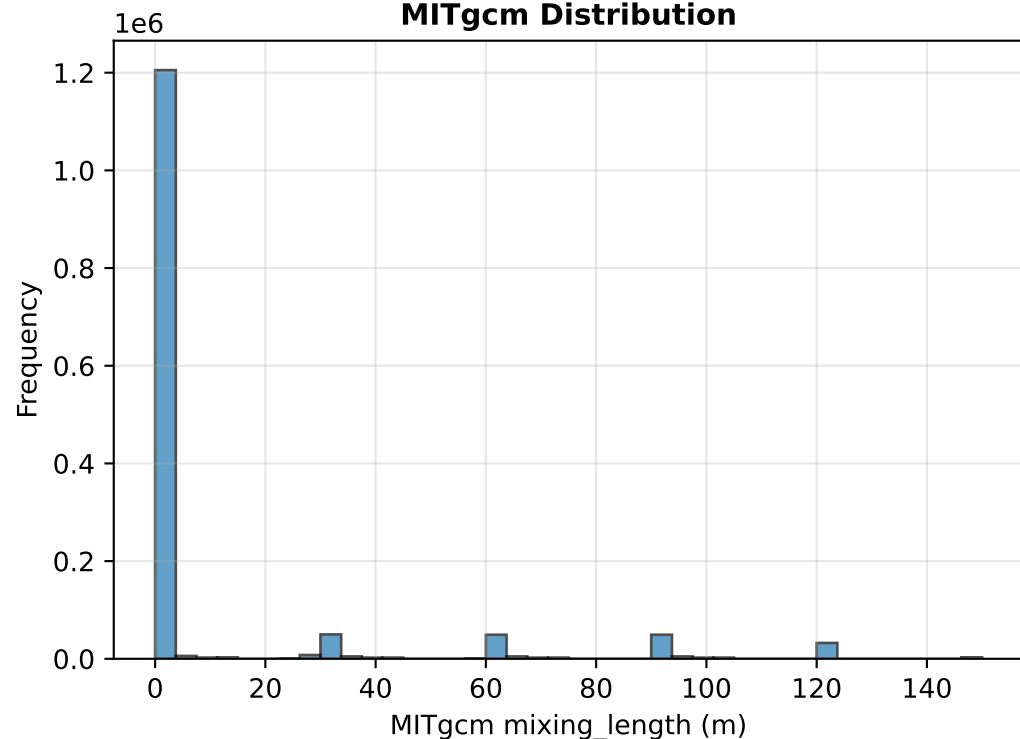
Median rel: 0.0000e+00
P95 rel: 0.0000e+00
Max rel: 3.0000e+00

Fraction >1% rel: 0.0770%
Points compared: 1437204
NaN-excluded: 362796

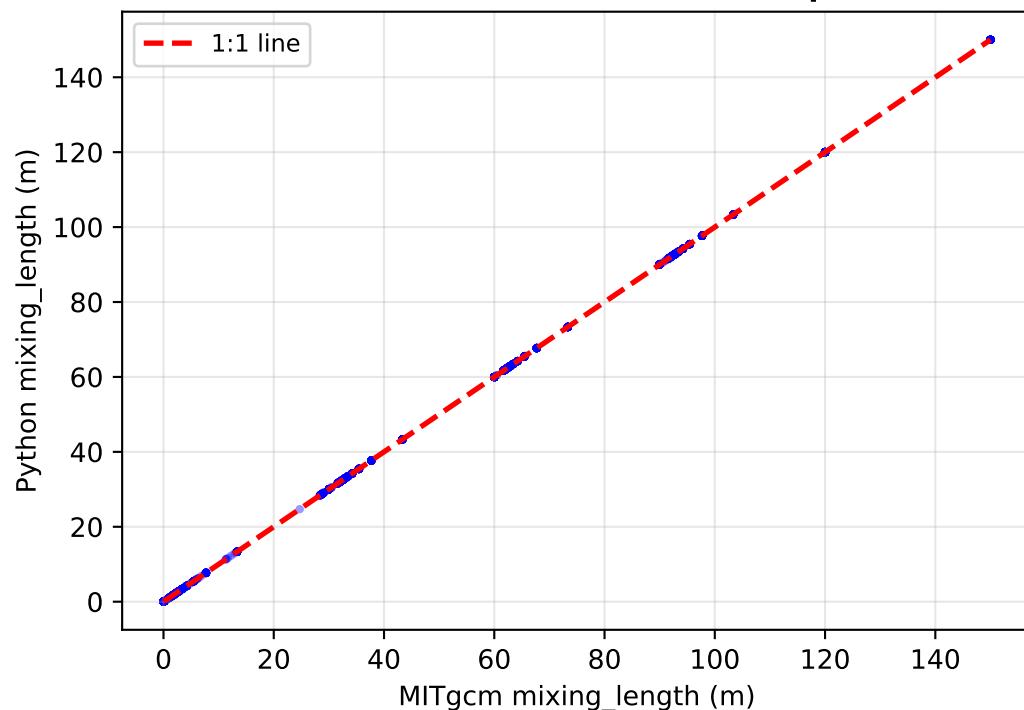
Assessment:
EXCELLENT

mixing_length (GGL90 mixing length (L)) -- MITgcm vs Python

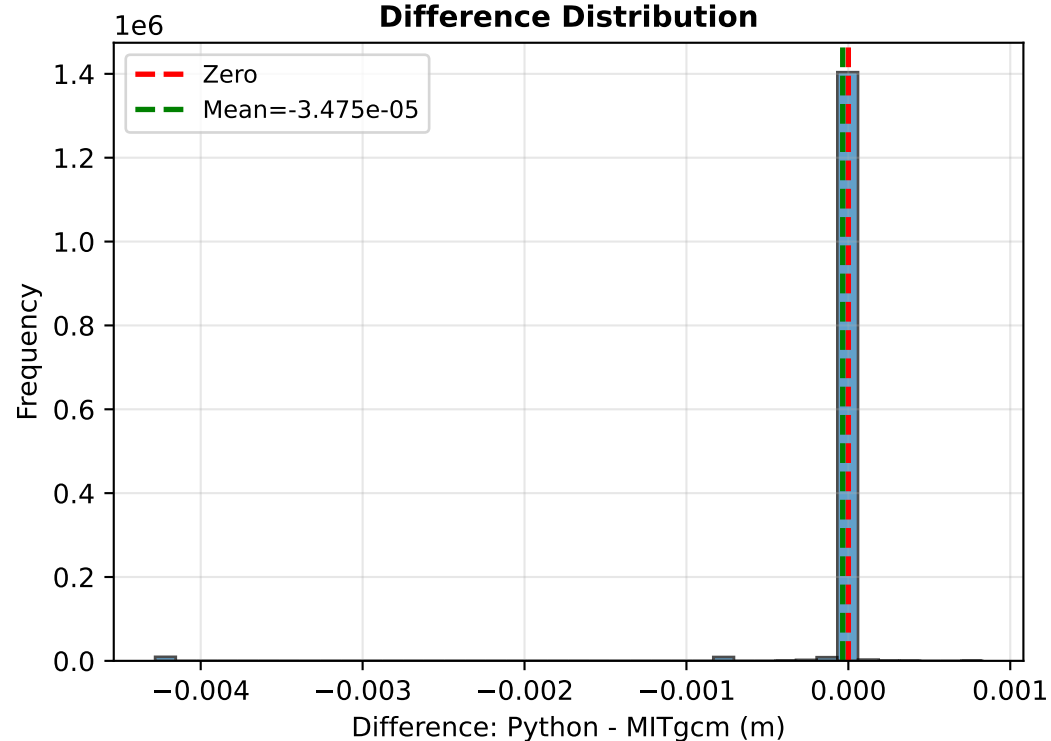
MITgcm Distribution



Scatter Plot (20000 of 1437204 pts)



Difference Distribution



mixing_length Statistics:

Median |diff|: 4.9873×10^{-18}
P95 |diff|: 2.1316×10^{-14}
Max |diff|: 4.2817×10^{-3}

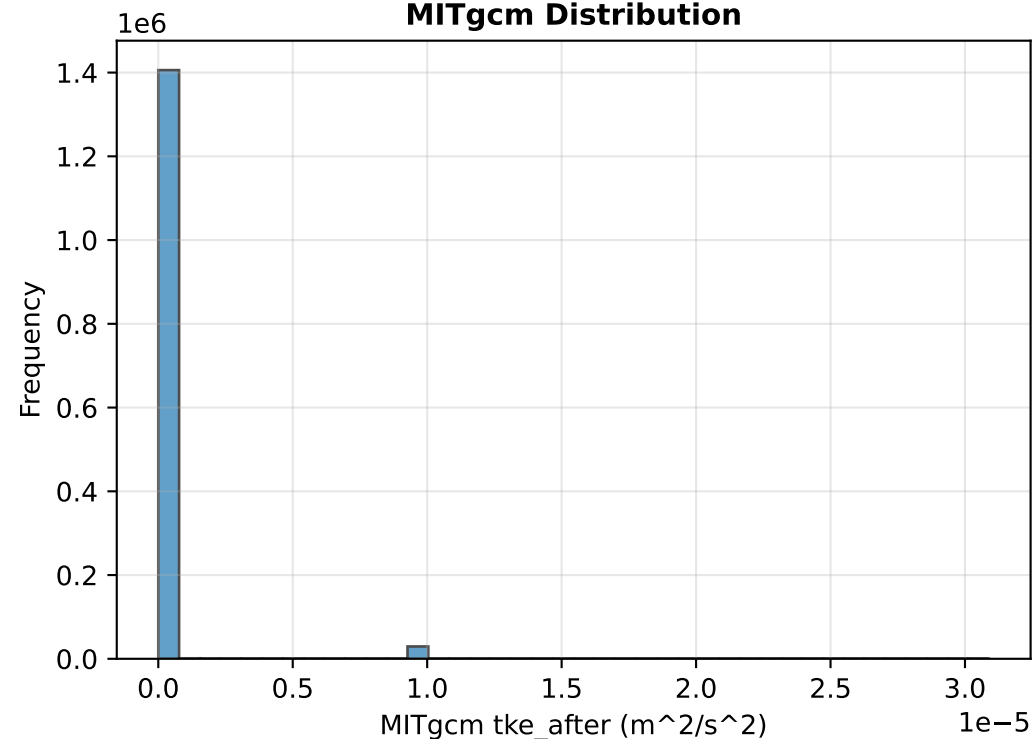
Median rel: 3.5691×10^{-15}
P95 rel: 6.4581×10^{-12}
Max rel: 3.2098×10^{-4}

Fraction >1% rel: 0.0000%
Points compared: 1437204
NaN-excluded: 362796

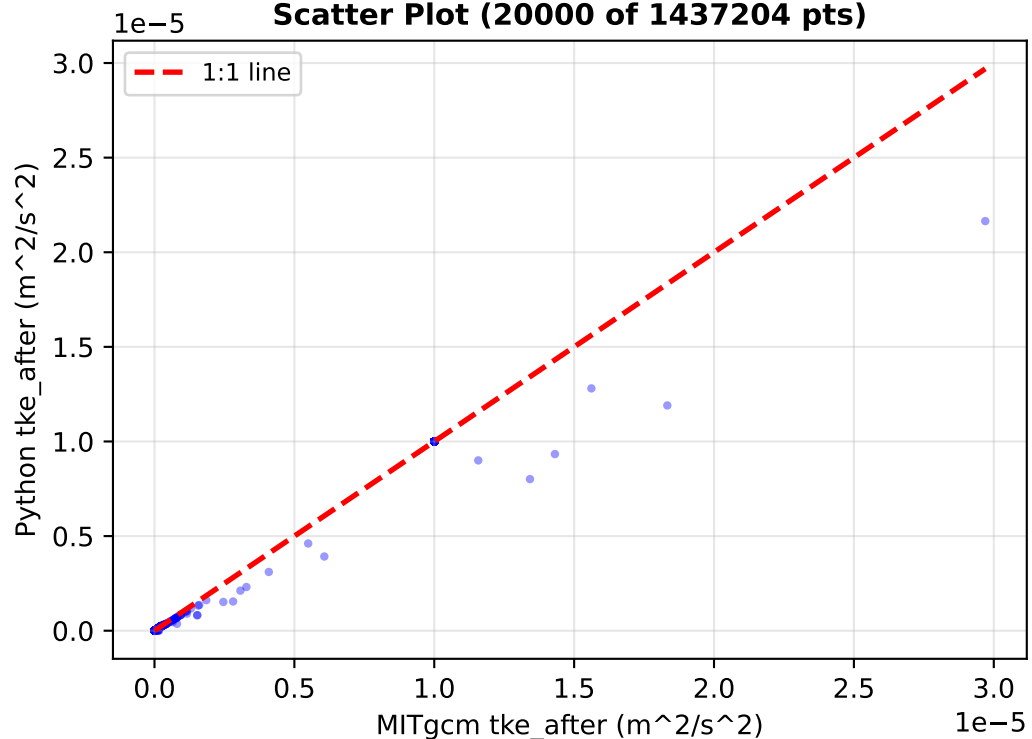
Assessment:
EXCELLENT

tke_after (TKE at t+1 (single-step replay)) -- MITgcm vs Python

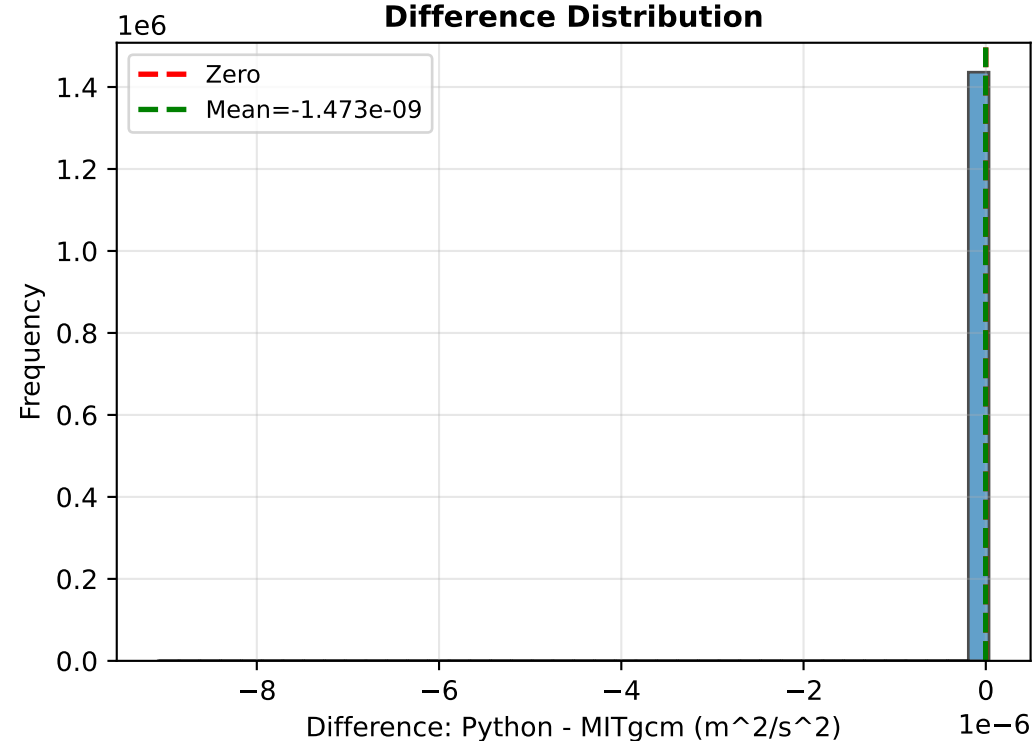
MITgcm Distribution



Scatter Plot (20000 of 1437204 pts)



Difference Distribution



tke_after Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 6.3151e-17
Max |diff|: 9.0811e-06

Median rel: 0.0000e+00
P95 rel: 1.3664e-06
Max rel: 2.3269e+00

Fraction >1% rel: 1.2478%
Points compared: 1437204
NaN-excluded: 362796

Assessment:
EXCELLENT

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ PASS

COMPONENT RESULTS:

visc_az: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 2.3898e-10
Fraction >1% rel: 0.0000%

diff_kz: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 2.9049e-03
Fraction >1% rel: 0.0770%

mixing_length: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 4.2817e-03
Fraction >1% rel: 0.0000%

tke_after: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 9.0811e-06
Fraction >1% rel: 1.2478%

PROVENANCE:

UUID Match: ☒ Yes
MITgcm UUID: c716d756-1f0d-4ab3-a7b5-d36c2ffa10e9
Python UUID: c716d756-1f0d-4ab3-a7b5-d36c2ffa10e9

ACCEPTANCE CRITERIA (mirrors generate_kpp_validation_report.py's own):

☒ EXCELLENT: median rel error < 0.1%
☒ GOOD: median rel error < 1.0%
☒ CHECK: median rel error >= 1.0% -- may be a real, already-characterized gap (see closed_issues.md), not necessarily a new defect; this report states the number, not the cause.